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Iteratively refined ChatGPT outperforms clinical mentors in generating high-quality interprofessional education clinical scenarios: a comparative study

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Abstract

Background Interprofessional education (IPE) is essential for promoting teamwork among healthcare professionals. However, its implementation is often hindered by the limited availability of interprofessional faculty and scheduling challenges in creating high-quality IPE scenarios. While AI tools like ChatGPT are increasingly being explored for this purpose, they have yet to demonstrate the ability to generate high-quality IPE scenarios, which remains a significant challenge. This study examines the effectiveness of GPT-4o, an advanced version of ChatGPT enhanced by novel methodologies, in overcoming these obstacles.

Methods This comparative study assessed clinical scenarios generated by GPT-4o using two strategies—standard prompt (a single-step scenario generation without iterative feedback) and iterative refinement (a multi-step, feedback-driven process)—against those crafted by clinical mentors. The iterative refinement method, inspired by actual clinical scenario development, employs a cyclical process of evaluation and refinement, closely mimicking discussions among professionals. Scenarios were evaluated for time efficiency and quality using the Interprofessional Quality Score (IQS), defined as the mean score assigned by multidisciplinary evaluators across five interprofessional criteria: clinical authenticity, team collaboration, educational alignment, appropriate challenge, and student engagement.

Results Scenarios developed using the iterative refinement strategy were completed significantly faster than those by clinical mentors and achieved higher or equivalent IQS. Notably, these scenarios matched or exceeded the quality of those created by humans, particularly in areas such as appropriate challenge and student engagement. Conversely, scenarios generated via the standard prompt method exhibited lower accuracy and various other deficiencies. Blinded attribution assessments by students further demonstrated that scenarios developed through iterative refinement were often indistinguishable from those created by human mentors.

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Conclusions Employing GPT-4o with iterative refinement and role-playing strategies produces clinical scenarios that, in some areas, exceed those developed by clinical mentors. This approach reduces the need for extensive faculty involvement, highlighting AI's potential to closely align with established educational frameworks and substantially enhance IPE, particularly in resource-constrained settings.

Keywords Interprofessional education, Clinical scenarios, ChatGPT, Artificial intelligence, Iterative refinement

Background

Interprofvering healthcare, ultimately improving patient care through collaboration [essional education (IPE) is recognized by the World Health Organization as a crucial teaching method to improve teamwork among healthcare professionals and bridge gaps in deli1, 2].

Despite its benefits, implementing IPE poses substantial challenges, particularly in the development of curriculum for educators [3]. Key challenges include engaging interprofessional faculty in curriculum development and coordinating their schedules, which is often difficult [4]. Virtual interprofessional education offers flexible online platforms that address some of these challenges [5, 6]. However, in less developed regions, the scarcity of diverse faculty continues to severely limit IPE integration. Moreover, creating appropriate clinical scenarios is crucial, as they allow students from various healthcare fields to collaboratively enhance their clinical and teamwork skills [7]. Nevertheless, developing these scenarios—which must be medically accurate, relevant, and suitably complex [8]—remains a time-consuming and significant barrier to advancing IPE.

In the context of medical education, ChatGPT has shown potential due to its broad knowledge base and ability to perform logical reasoning, demonstrating utility in various tasks such as generating multiple-choice questions [9], assisting with teaching evaluations [10], and supporting academic writing [11–13]. The application of ChatGPT in creating clinical scenarios for IPE holds promise for improving efficiency. However, this task typically requires the collaborative efforts of interprofessional faculty, each contributing unique expertise. As such, creating clinical scenarios for IPE is a more complex challenge than generating multiple-choice questions or writing summaries and evaluations, as highlighted in previous studies.

To date, there are no published studies that demonstrate the successful creation of high-quality clinical scenarios using large language models. Developing IPE clinical scenarios that align with the World Health Organization Framework for IPE remains a challenging task. Inspired by the concept of agentic workflows in ChatGPT [14, 15], we recognize that generating IPE cases requires the involvement of multiple experts playing different roles, providing diverse perspectives, and engaging in discussions and iterative modifications [16]. This mirrors the iterative refinement and role-playing strategies

employed by ChatGPT [17, 18]. By incorporating these strategies, the quality of ChatGPT-generated clinical scenarios is significantly improved compared to traditional prompting methods. In this study, we are among the first to report on the practical method of generating IPE clinical scenarios using artificial intelligence. Rigorous assessment of the efficiency and quality of these scenarios provides crucial evidence supporting their integration into curriculum development, offering a novel and practical solution to current IPE challenges.

Methods

Study-defined standards for creating multidisciplinary clinical scenarios

Both clinical mentors and ChatGPT-generated scenarios encompass four key types: acute and emergency care, chronic and primary care, community and public health, and specialized and supportive care. For each type, eight distinct scenarios are required. This standard ensures a comprehensive evaluation of whether the scenarios—whether created by GPT-4o or clinical mentors—meet the diverse educational needs across various teaching settings. Furthermore, each clinical scenario must incorporate the same four specialties to ensure comprehensive educational coverage, as outlined in eTable 1 in Supplementary Material 1.

Creation of clinical scenarios by multidisciplinary clinical mentors

Multidisciplinary clinical mentors are recruited from among residency program instructors and consist of clinical mentors from various relevant specialties. These mentors collaborate both online and face-to-face to discuss and refine scenarios, ultimately crafting scenarios that meet specified IPE requirements.

Creation of clinical scenarios by GPT-4o

All clinical scenarios generated by ChatGPT use GPT-4o, the latest and most advanced language model from OpenAI, available through a free-tier access. The standard prompt method generates scenarios based on the following input, without iterative modifications (detailed in eAppendix 2 of Supplementary Material 1):

Create an interprofessional clinical scenario in [specified type], involving students from [specified specialties]. The scenario should emphasize team-

work, with each specialty assuming distinct roles such as diagnosis, medication management, direct care, patient education, care coordination, and health promotion. Focus on patient-centered care, effective communication, and interdisciplinary collaboration, in alignment with the educational goals of real-world interprofessional practice.

We also utilize a specialized iterative refinement method to enhance realism, accuracy, and interdisciplinary integration of clinical scenarios through a structured three-step process. The process begins with gathering and synthesizing clinical data from multiple specialties to create a preliminary clinical scenario that reflects the perspectives and needs of each discipline involved. A comprehensive analysis of relevant medical history, physical examinations, and preliminary tests is conducted to ensure the scenario provides a holistic view of the case. The preliminary clinical scenario generated in Step 1 often lacks essential case details, is not sufficiently accurate, and typically does not encompass all specialties, making it insufficient for meeting IPE requirements. In the second phase, GPT-4o simulates students from various specialties who evaluate and refine the preliminary scenario, ensuring it covers all critical aspects of diagnosis, treatment, and discipline-specific details such as primary symptoms, diagnostic tests, medical history, and treatment strategies. In the final phase, GPT-4o simulates experts from the specified specialties, further refining the scenario to improve its realism and accuracy. Experts enhance details related to symptoms, physical examinations, and diagnostic tests pertinent to diagnosis and treatment, re-evaluating the case to ensure its medical accuracy and preparing it for educational use. Details

of the multi-step iterative refinement and role-playing strategy, along with examples of prompts, are provided in Table 1(detailed in eAppendix 3 of Supplementary Material 1).

Each AI-generated scenario is initially reviewed and approved by a mentor outside the specified specialties. In contrast, clinical scenarios developed by multidisciplinary expert teams are refined through discussions among clinical mentors from the relevant specialties.

Blinded assessments by multidisciplinary clinical experts and students

All clinical scenarios were randomly distributed and anonymously evaluated to ensure that reviewers remained unaware of each scenario’s origin, thus preserving objectivity. The assessment criteria included clinical authenticity, team collaboration, educational goals, appropriate challenge, and an additional criterion of student engagement (detailed in eAppendix 1 of Supplementary Material 1).

Clinical experts and student evaluators were recruited from The Second Xiangya Hospital of Central South University. Clinical experts, invited from relevant specialties, were senior physicians who did not participate in the initial creation of clinical scenarios and voluntarily agreed to participate. They were responsible for evaluating the first four criteria. Student evaluators were recruited from residents at our institution in specialties relevant to the scenarios, all of whom had prior experience with IPE. All participants provided written informed consent prior to evaluation.

Before scoring, both experts and students underwent standardized training on the assessment criteria to ensure consistency. Each scenario was reviewed by evaluators

Table 1 Iterative refinement and role-playing instructions with examples of prompts

Step description	Prompt example
Step 1 Analytical preparation and preliminary case scenario formulation Gather and synthesize clinical data from multiple specialties to formulate a preliminary clinical scenario that reflects the perspectives and needs of each discipline involved.	Generate an interprofessional education scenario for [specified scenario type], involving the following specialties: [specified specialties]. Conduct a thorough analysis of each specialty's relevant medical history, physical exams, and preliminary tests. Integrate these findings into a cohesive and detailed scenario. The scenario should be organized as follows: background and scenario introduction (detailed description), initial assessment and data (detailed description), perspectives of different specialties (brief summary), collaborative professional tasks (brief summary).
Step 2 Student perspective evaluation and refinement Students from various specialties refine the scenario, ensuring it covers all relevant aspects of diagnosis and treatment.	Simulate the perspectives of students from each of the [specified specialties] to evaluate and refine the preliminary clinical scenario. Step 1: Assess and adjust the scenario to ensure comprehensive coverage of key symptoms, diagnostic evaluations, relevant medical history, and treatment strategies across all disciplines. Step 2: Enhance the scenario by adding detailed descriptions of symptoms, physical examinations, and diagnostic tests essential for accurate diagnosis and treatment, thereby improving its realism. The output format should match that of the preliminary clinical scenario. The preliminary clinical scenario is as follows: [...]
Step 3 Expert perspective evaluation and refinement Experts refine the scenario to improve its realism and accuracy, finalizing it for educational use.	Simulate the perspectives of experts from each of the [specified specialties]. Step 1: Enhance the scenario by refining details related to symptoms, physical examinations, and diagnostic tests essential for accurate diagnosis and treatment, maximizing its realism. Step 2: Re-evaluate the scenario for medical accuracy and output the final refined version. The output format should match that of the preliminary clinical scenario. The preliminary clinical scenario is as follows: [...]

from four distinct healthcare disciplines. Scoring was performed using a 5-point Likert scale, with details provided in Supplementary Material 1. The Interprofessional Quality Score (IQS), as defined in our study, was calculated as the average of scores provided by evaluators from diverse disciplines across all assessment criteria, thereby offering a comprehensive evaluation of scenario quality. The Interprofessional Quality Score (IQS) is defined as the mean score assigned by evaluators from relevant specialties across five interprofessional assessment criteria—clinical authenticity, team collaboration, educational alignment, appropriate challenge, and student engagement—that were specifically developed for this research, as currently there are no validated scoring methods or criteria available for evaluating IPE scenarios (Fig. 1).

Blinded source attribution by students

Following the assessment of student engagement, the same group of student evaluators was tasked with identifying whether the scenarios were generated by AI or by human mentors. Importantly, the students were blinded to the total number of scenarios produced by each group, ensuring an unbiased and objective assessment process.

Statistical analysis

We performed independent samples t-tests to compare the mean Interprofessional Quality Score between GPT-4o-generated and human-generated clinical scenarios. A χ^2 test was utilized to compare attribution frequencies (Human vs. AI) across various scenario generation methods. Results with a p value of less than 0.05 were considered statistically significant. All statistical analyses were conducted using SPSS software (Version 21).

Results

Creation of clinical scenarios

Adhering to the standards for multidisciplinary clinical scenarios, each of the groups—clinical mentors, standard

prompt, and iterative refinement—developed a total of 32 scenarios across four specific types. The average time required to create each scenario was 118 ± 23 min for clinical mentors, 4 ± 2 min for standard prompt, and 9 ± 2 min for iterative refinement, respectively (see Fig. 2a). Clearly, the scenarios generated by GPT-4o are produced in significantly less time than those crafted by human mentors. Furthermore, even when employing strategies such as iterative refinement and role-playing to guide the generation process with GPT-4o, the time required remains reasonable.

Evaluation of scenario quality

The IQS is a comprehensive metric used to assess the quality of clinical scenarios across various dimensions. Scenarios generated using the standard prompt method by GPT-4o consistently exhibit significantly lower IQS than those crafted by clinical mentors, especially in key areas such as clinical authenticity (3.50 ± 0.35 vs. 4.00 ± 0.35 ; $p < .01$), team collaboration (3.27 ± 0.28 vs. 3.66 ± 0.56 ; $p < .01$), and student engagement (3.01 ± 0.27 vs. 3.49 ± 0.43 ; $p < .01$). These results suggest that the quality of scenarios generated through the standard prompt method remains suboptimal.

Surprisingly, scenarios generated using the iterative refinement method demonstrated notable improvements. They achieved higher IQS than those created by clinical mentors in categories such as appropriate challenge (3.61 ± 0.49 vs. 2.98 ± 0.34 ; $p < .01$) and student engagement (3.88 ± 0.57 vs. 3.49 ± 0.43 ; $p < .01$), while attaining comparable quality ratings in clinical authenticity and three other criteria (see Fig. 2b). Compared to the standard prompt method, the iterative refinement method also showed significant improvements in generating high-quality scenarios, as evidenced by improvements across all evaluated dimensions.

This result underscores the considerable advantages of the iterative refinement method, highlighting its

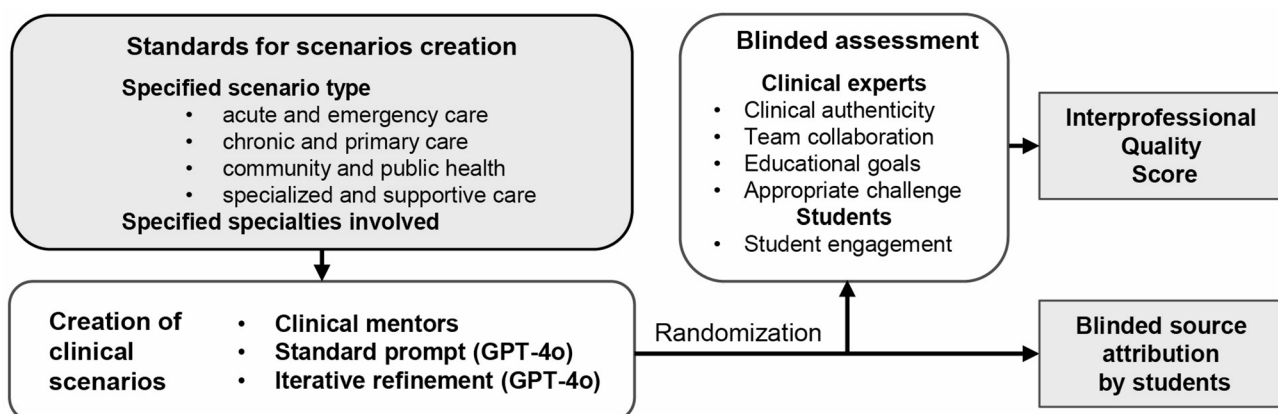


Fig. 1 Schematic representation of the study design

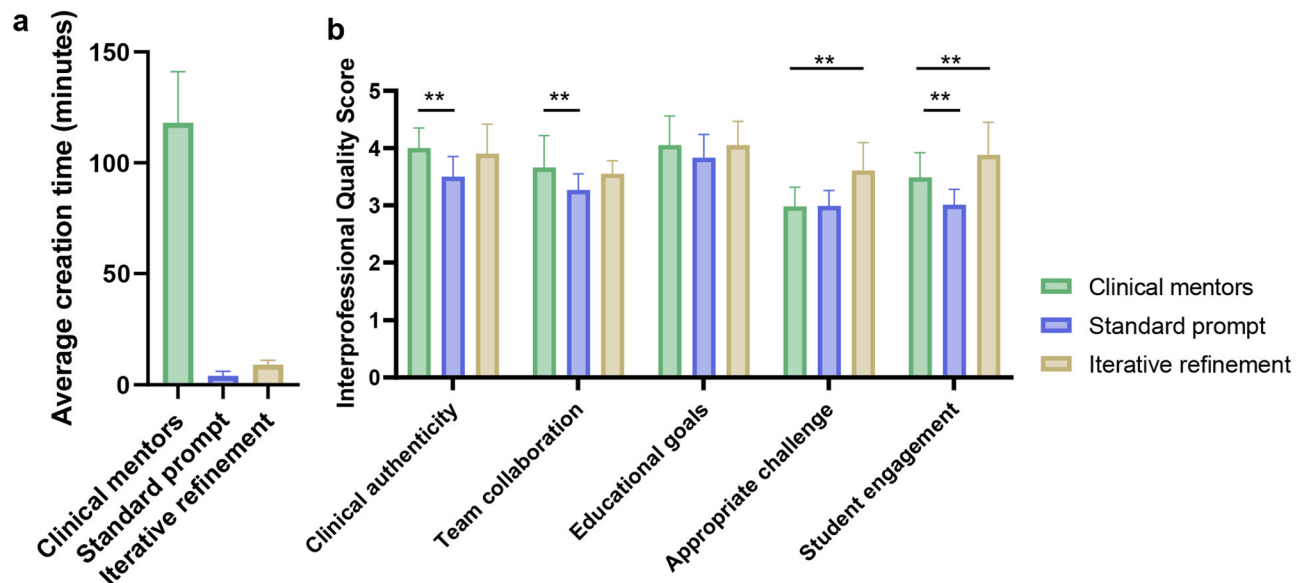


Fig. 2 Time and quality comparison results of clinical scenarios generated by GPT-4o under different strategies versus human creation. **a)** Average time spent by different groups to create clinical scenarios, with the time for the clinical mentors group representing the cumulative effort of all involved multidisciplinary mentors. **b)** Comparison of IQS results across various metrics for clinical scenarios created by different methods. Statistical significance was determined using two-tailed T-tests, with * indicating $p < .05$, ** indicating $p < .01$ and *** indicating $p < .001$

Table 2 Results of source attribution assessment

	Blinded source attribution		p Value (vs. clinical mentors)
	Human	AI	
Clinical mentors(N= 32)	19	13	NA
Standard prompt(N= 32)	8	24	0.01
Iterative refinement(N= 32)	16	16	0.61

effectiveness in generating high-quality clinical scenarios that not only meet or exceed the standards set by experienced clinical mentors but also address the limitations of the standard prompt approach. This evidence supports iterative refinement as a robust strategy for producing scenarios of superior quality.

Source attribution assessment

Scenarios generated using the standard prompt method by GPT-4o are more likely to be identified as AI-generated, whereas those created using the iterative refinement method are perceived similarly to those crafted by clinical mentors (Table 2).

Discussion

Generating high-quality IPE clinical scenarios using artificial intelligence presents significant challenges but can substantially alleviate the difficulties associated with the scarcity of interprofessional faculty resources. However, studies, including those by Cheung and Lau et al. [9], indicate that ChatGPT struggles to generate effective scenarios using standard prompts. This shortcoming likely

arises from the complexity and expertise required to create contextually relevant scenarios.

In blind evaluations conducted by clinical experts and students, scenarios generated using standard prompt method exhibited lower accuracy and multiple deficiencies compared to those crafted by clinical mentors. Our study applied iterative refinement and role-playing with structured prompts innovatively to enhance scenario generation. This approach enabled the scenarios developed through iterative refinement to achieve quality on par with, and in aspects such as appropriate challenge and student engagement, even exceed the level of those created by human instructors.

This improvement is attributed to GPT-4o's ability to iteratively refine cases, tailoring them to the specific needs of each discipline. Iterative refinement outperforms standard prompts by employing a structured, stepwise approach that guides the model through progressive adjustments [17, 19]. This process begins with task decomposition, breaking complex scenarios into sequential steps, with each prompt focusing on a particular aspect of the output. For example, initial prompts establish a foundational framework, while subsequent

prompts refine details like clinical accuracy and inter-professional roles. Each iteration incorporates feedback, helping the model identify and correct errors, minimize hallucinations, and enhance reliability. Moreover, iterative prompts enable dynamic adaptation to changing task requirements, ensuring content remains contextually relevant and aligned with multidisciplinary perspectives. By systematically guiding the model with focused, feedback-driven prompts, iterative refinement maximizes GPT-4o's capabilities, producing more accurate, reliable, and contextually appropriate outputs.

This method generates high-quality scenarios without direct input from multidisciplinary instructors, addressing key challenges in IPE such as faculty shortages and scheduling conflicts [4, 20]. It is particularly beneficial in underdeveloped regions or lower-tier teaching hospitals, where limited faculty often hinder IPE implementation [21, 22]. Supported by an AI platform, this approach enhances scalability and accessibility, expanding its application across diverse educational environments. However, despite the high quality of AI-generated scenarios, instructor review and potential modifications are necessary to ensure their effectiveness and relevance in educational settings.

Our assessment relied on subjective scoring by evaluators, which, despite standardized training, may introduce variability. Additionally, our study did not include formal inter-rater reliability testing, representing a limitation. Future studies should incorporate such analyses to further enhance the validity and reproducibility of the IQS method.

While the iterative refinement strategy significantly improves scenario quality, the potential benefits of additional iterations (repeating Steps 2 and 3) in further reducing errors and enhancing detail have not been explored in this study. Scenarios generated using the standard prompt method often oversimplified inter-disciplinary roles and lacked sufficient clinical detail, especially in diagnostic reasoning. Although these deficiencies were reduced through iterative refinement, they were not fully eliminated.

To further validate the quality of GPT-4o-generated materials, future integration into IPE practices, coupled with assessments of student engagement and learning outcomes (both qualitative and quantitative), could provide valuable empirical evidence and enhance teaching effectiveness. Future research could investigate the integration of AI tools with faculty oversight to mitigate potential inaccuracies and produce scenarios that more closely align with IPE objectives. Such collaboration has the potential to improve the contextual accuracy and educational relevance of AI-generated materials, paving the way for advancing IPE practices and AI-assisted education.

Conclusions

Using the advanced GPT-4o, scenarios generated by the standard prompt method continue to exhibit various shortcomings. Furthermore, employing strategies such as iterative refinement and role-playing allows GPT-4o-generated clinical scenarios to match or even exceed those developed by clinical mentors. Applying these strategies can achieve significant time savings without the need for extensive interprofessional faculty involvement. Nevertheless, a careful review by instructors remains essential. This approach not only showcases the substantial potential of AI in creating personalized learning materials but also presents an innovative and effective solution to the current challenges in IPE.

Abbreviations

IPE	Interprofessional education
IQS	Interprofessional Quality Score
AI	Artificial intelligence

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12909-025-07414-1>.

Supplementary Material 1

Acknowledgements

Not applicable.

Author contributions

Tian Qingquan: data collection and writing of the manuscript; Liu Ganglei, Zhou Jinyu, Zheng Yanwen, Zhang Zequn and Ren Feng: data collection and organization; Zou Bin: data analysis; Wang Qiyuan: study design, recruitment of faculty and students for the study; Wang Shalong: revision of the manuscript, organization and providing suggestions.

Funding

This work was supported by Young Researcher Fund Project of Ministry of Education Foundation on Humanities and Social Sciences under Grant 24YJC880135 and the Degree & Postgraduate Education Reform Project of Central South University under Grant general project-54.

Data availability

The datasets during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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Received: 25 June 2024 / Accepted: 28 May 2025

Published online: 05 June 2025

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